A Medical Video Coding Scheme with Super Resolution Reconstruction

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***Abstract*—Medical video is increasingly playing a vital role in medicine, with its transmission becoming more mainstream. To support this, better compression techniques which can maintain the diagnostic capabilities of the video are required, especially for transmission over bandwidth-limited channels. This work presents a novel medical video compression system that uses Super Resolution (SR) as a processing step and an Enhancement Layer (EL) to enhance the quality in Regions of Interest (ROIs). This system achieves an average gain of 28.25% in Compression Ratio (CR) over full resolution lossless compression, at a Peak Signal-to-Noise Ratio (PSNR) of 38.17dB, a Structure Similarity (SSIM) of 0.9843 and Multi-Scale Structure Similarity (MS-SSIM) of 0.9991. The encoding time is reduced due to downscaling of the medical video. Subjective evaluations show that the diagnostic quality is retained.**

***Index Terms*—medical video compression, super resolution**

I. Introduction

Advances in technology are creating a growing demand for mobile health (mHealth) systems, resulting in increasing needs to transmit medical video, both for educational and diagnostic purposes. Moreover, CT and MRI scans can be treated as video data due to the small incremental changes between slices. In these modalities, improvements in spatial resolutions, reductions in slice thickness and increases in bit depths are leading to larger file sizes. Furthermore, population screening for early diagnosis of certain pathologies using these modalities creates a large amount of data that needs to be stored or transmitted. This highlights the need for efficient compression systems of such medical videos and data.

This work proposes a *diagnostically* lossless medical video compression scheme. Such systems need not to be fully lossless, but should allow the same diagnostic quality as the original video. The proposed system downscales the video prior to compression, allowing for an additional compression gain. To restore the original resolution, Super Resolution (SR) is performed using a Convolutional Neural Network (CNN). Since downscaling reduces the quality of the data, an Enhancement Layer (EL) is also transmitted to improve the quality in the diagnostic Regions of Interest (ROIs), allowing these regions to be lossless or visually-lossless.

The research work disclosed in this publication is funded by the ENDEAVOUR Scholarships Scheme

The rest of the paper is organized as follows: Section II gives a background on medical video compression methods and SR. Section III discusses the implementation of the compression system, with Section IV discussing in more detail the training of the SR CNN. Section V presents the results and final remarks are given in Section VI.

II. Related Work

A. Medical Video Compression

Various attempts at efficiently compressing medical video are found in literature. Medical video has particular requirements as it is used for diagnostic purposes. Thus, noticeable degradations in diagnostically important ROIs are unacceptable, as these can lead to misdiagnosis.

Although works which use H.264 codecs give encouraging results, [1] confirms that the newer H.265/High Efficiency Video Coding (HEVC) standard is superior, even for medical video. This was also confirmed by clinical evaluation.

In [2], Chang *et al.* improve lossless HEVC compression by catering for the inefficiencies caused by sharp edges and textures present in medical video. To this aim, they make use of geometry to predict missing pixels in partially reconstructed blocks, allowing them to be considered as candidates for motion vector prediction. Guarda *et al.* [3] improve the predictor for lossless HEVC, where Differential Pulse Code Modulation (DPCM) is first used on adjacent frames and then Least Squares Prediction (LSP) is used as a predictor. LSP makes use of a linear adaptive filter which is optimized based on a training window, which is re-defined for every pixel to be predicted and whose support is also extended temporally.

Whilst the previous methods are based on HEVC, other attempts that use non-standard methods exist. In [4], medical data is compressed losslessly through a quadtree structure which decomposes the pixel-wise difference between successive frames. Predictors optimized for each block in the quadtree are used to compress this data, using reference pixels which can also be found in previous frames. Further compression is achieved by context modelling and arithmetic coding. In [5], a near-lossless technique is developed which is similar to HEVC but uses smaller block sizes and the Lossless Hadamard Transform, with block matching also including transformations of blocks, such as flips.

Chandrika *et al.* [6] use a Just Noticeable Distortion (JND) profile, which gives a threshold below which errors are unnoticeable by the Human Visual System (HVS). Quantizing is based on the value obtained by the JND profile and a quantization weighting factor. This is followed by symmetry detection to improve block matching. The 3D wavelet scheme proposed in [7] results in two sets of frames: one containing the average signal and the other containing details. These details are mainly generated by movement and can be discarded. This is followed by entropy coding using a 3D conscious run-length coding scheme, followed by arithmetic coding.

B. Super Resolution

Super Resolution (SR) aims to answer the question of how a High Resolution (HR) image can be obtained from a Low Resolution (LR) input. Whilst basic interpolation methods can be used, their results tend to be blurry. In literature, techniques focusing on both single-frame and multi-frame SR can be found, with multi-frame techniques gathering traction for use in video SR. Learning-based methods are the most popular methods that are currently being researched. They can be adapted to both single and multiple frames. Here the focus is on multi-frame SR in the context of video, specifically learning-based methods which make use of neural networks, as they are fast and require no dictionary at the receiving end, whilst managing to achieve very good upscaling results.

Inspired by [8], Kappeler *et al.* [9] construct a CNN video SR model that uses motion compensated neighbouring frames to improve SR performance. A different approach is taken by Li *et al.* [10], where the method resolves high-frequency residual details between an image upscaled by bicubic interpolation and the original HR frame. Similar to [9], motion compensated neighboring frames are used to improve results.

Whilst both [9], [10] require bicubic interpolated inputs, in newer methods which tackle single image SR [11], [12] the resolution is increased at the end of the network using a deconvolution layer. In [13], Caballero *et al.* use this technique whilst also making use of neighboring frames, using a spatial transformer network to perform motion compensation.

III. Method

A. Algorithm Overview

Figure 1 shows a high-level representation of the proposed system. The spatial resolution of the frames is reduced at the transmitter to save bandwidth. At the receiver, rescaling back to the native resolution is done via an SR CNN. Medical videos require that specific ROIs have high quality. Thus, an EL is transmitted with each frame to enhance these regions.

B. The Video Encoder

The standard HEVC video encoder, obtained from [14], is used in this work. The encoder was configured using the low delay profile and the group of pictures was set to 12. The videos were compressed both lossless and lossily.

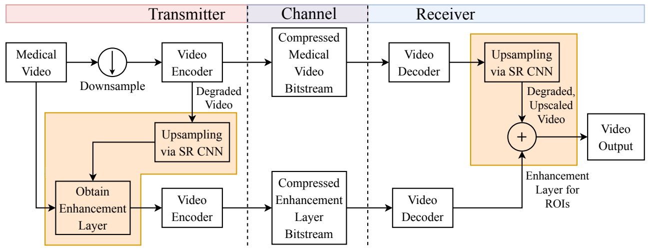


Fig. 1: The proposed method, contributions shown in orange.

C. The Super Resolution Convolution Neural Network

Figure 2 shows the architecture used in this work. The model takes two inputs: the LR frame that will be upscaled at time t and the motion compensated LR frame at time $t-1$.

The next two layers, perform feature extraction and shrinking, similar to [11]. The shrinking layer reduces the dimensionality of the feature maps, resulting in an overall reduction in computation time. The same amount of filters and filter sizes as in [11] are used. The next layer fuses together the shrunken feature maps of the two inputs. Fusion is done at this stage since the work in [13] shows that this is the most efficient, both in terms of speed and quality. This is followed by M convolutional layers which perform mapping onto the vectors which will represent the HR frame. Next is an expanding layer, which can be seen as the inverse of the shrinking layer.

The final deconvolution layer performs the actual upscaling. It is composed of a single filter, whose padding, kernel size and stride are carefully chosen to ensure that the output is *exactly* twice or four times the input size, depending on the original downscaling factor. More information on parameter selections can be found in [15].

After all convolutional layers, a bias term is added, followed by a Parametric Rectified Linear Unit (PReLU) [16] activation function. The architecture was implemented in Python, using Keras version 2.1.4 with TensorFlow version 1.5.0 backend.

D. The Enhancement Layer

An EL is transmitted alongside the medical video to correct for losses in the video caused by downscaling and lossy compression (if a Quantization Parameter (QP) > 0 is used). The EL allows details to be retained at diagnostic ROIs, allowing for correct diagnosis to take place.

At the transmitter, the EL is obtained by using the SR CNN to upscale the video, and subtracting the result from the original video. This is done only for ROIs. At the receiver, this residual is added to the ROIs to improve their quality.

In this work, the ROIs are considered as rectangular regions with a static size and location throughout the whole video. Thus, the ELs can be considered as frames, allowing them to be compressed losslessly via HEVC. This requires that the top left coordinate of the region is also transmitted, adding an extra 2 bytes overhead. For 8 bit medical video, the residual requires 9 bits. However, HEVC does not support 9 bit compression. To cater for this, residuals were saturated to the range [-128, 127], which is the range where most errors are expected to be, and then shifted to the range [0, 255].

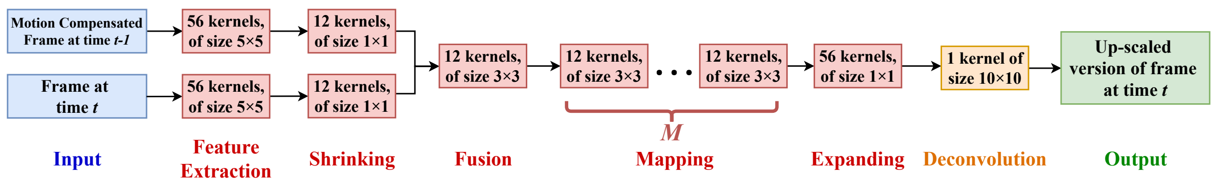


Fig. 2: The proposed CNN architecture. Red layers are convolutional layers; orange layer is the final deconvolutional layer.

IV. Training

A. Data

Medical data composed of CT, MRI and Ultrasound scans was obtained from [17] and [18]. To perform Motion Compensation (MC), OpenCV version 3.4.0's CPU based Dense Inverse Search optical flow algorithm with the *Medium* preset was used. A ready-available MC algorithm was used since its development was not the scope of this work.

The total data amounted to 60,000 frames. This was split in a 60:20:20 ratio to form the training, validation and test sets. An intermediate test set consisting of approximately 6,000 frames was created for a specific purpose, as discussed in Section IV-B. Each set had an almost equal amount of CT, MRI and Ultrasound content of various body parts.

The training and validation data was split into 64×64 patches. This resulted in a lot of patches which had little to no texture, contributing minimal information to the training process. Such patches were removed leaving a total of 1,414,656 training and 471,552 validation patches.

OpenCV's Bicubic interpolation method was used to downscale the data used for training, validation and testing.

B. Cost Function

Although most works default to the ℓ_2 norm as a cost function, Zhao *et al.* [19], show that a mix of Multi-Scale Structure Similarity (MS-SSIM) and ℓ_1 loss functions is better for SR purposes. Thus, the cost function is defined as follows:

$$J(w) = \beta \cdot J^{MS-SSIM}(w) + (1-\beta) \cdot J^{\ell_1}(w) + 1e^{-5}\lambda||w||^2 \quad (1)$$

where β acts as a weighting factor and $\lambda||w||^2$ is used for regularization. A λ value of $1e^{-5}$ is used. This is a slight variation of the function in [19], since there is no mathematical or quantifiable justification for the inclusion of the Hadamard multiplication in [19].

To determine β, the CNN with $M = 8$ was trained for β values between 0.6 and 0.9 in steps of 0.1. Given that the loss function is varied for every network, the performance of the networks cannot be determined by comparing their validation losses. Thus, the Structure Similarity (SSIM) and the Peak Signal-to-Noise Ratio (PSNR) values were also output. From these values, it was noted that, although all β values result in almost equally capable networks, the model with β set to 0.6 outperformed the others. A test was performed on the uncompressed intermediate test set described in Section IV-A, which confirmed that setting β to 0.6 gives the best results.

C. Other Training Details

The CNN was trained for 3 different values for M: 6, 8 and 10, for both upscaling factors of 2 and 4, with a mini-batch size of 128. The Adam [20] optimiser was used, which computes individual learning rates (α values) for different parameters.

The filters' values for the $\times 2$ upscaling architectures were randomly initialized from a Gaussian distribution having $\mu = 0$ and σ based on the number of filters in the layer and the initial value of the PReLU function parameter [16]. The biases and the PReLU function's initial value were set to 0.

For the $\times 4$ upscaling models, two options were considered:

1) Random initialization, as done for $\times 2$ upscaling
2) Initialization with the converged weights and biases of the $\times 2$ CNNs, inspired by [11].

Both initialization methods were tested on the model with $M = 6$, allowing all layers to be trainable. Initialization using the second method allowed the model to converge faster and achieve better validation loss results. For this reason, the architectures with $M = 8$ and $M = 10$ were also initialized with the converged $\times 2$ upscaling values prior to training.

V. Results

This section discusses the results achieved by the proposed compression method when the downscaled medical video was compressed losslessly and with QP values between 24 and 32 via HEVC. Upscaling via bicubic interpolation acts as a reference to SR upscaling. Two scaling factors are considered: 2 and 4. To have a compression baseline, the full resolution video is also compressed with HEVC. The EL is always compressed losslessly at its native resolution with HEVC.

A. Objective Quality Metrics

The PSNR, SSIM and MS-SSIM give an indication of the visual quality of the compressed data. A graphical comparison between the objective quality metrics obtained by full resolution and the $\times 2$ upscaling methods is given in Figures 3(a), (b) and (c). The $\times 4$ methods' results are not shown, however they follow similar trends although at lower quality values. It was observed that changing the parameter M of the CNNs had little to no effect on these objective quality metrics, which might mean that the CNN is already operating at its optimal point. For this reason, only the results for $M = 10$ are shown.

In terms of PSNR, it is clear that the CNNs outperform bicubic interpolation for all QP values. However, as the QP values increase, the gain that the CNNs provide diminishes. The CNNs also give a larger improvement over bicubic interpolation when upscaling by 2 rather than by 4. The EL has a larger effect when upscaling via the CNNs than when using bicubic interpolation, with the largest increase observed when full resolution compression takes place.

Similar observations can be made for the SSIM, as seen in Figure 3(b). This suggests that the CNNs are capable of preserving structure better. In contrast to the PSNR, the CNNs' gain over bicubic interpolation is larger for $\times 4$ than for $\times 2$ upscaling. Again contrary to the PSNR, the EL gain is largest for bicubic interpolation and this gain increases with QP.

For the MS-SSIM, the results have similar trends to the SSIM. [21] states that for values of ≈ 0.99 or higher, distortions present are not visually noticeable. Figure 3(c), shows that, without the EL, all $\times 2$ upscaling methods surpass this value *only* when lossless compression of the medical video takes place. With the EL, this goes up to a QP of 26. In all these cases, however, the CNNs surpass this threshold by a larger value than bicubic interpolation. For $\times 4$ upscaling, none of the methods achieve a good enough MS-SSIM value without the EL. With the EL this threshold is only exceeded by the losslessly compressed CNN based methods.

B. *Subjective Quality Metrics*

Subjective testing was performed to assess whether the compressed video can be used for diagnosis. This was preformed according to the P.910 ITU-T standard [22] using a 5 level scale, where 1 represents bad quality and 5 represents excellent quality. 8 experts, including radiologists and consultants, were recruited. The participants were shown 3 videos (a CT, MRI and Ultrasound scan), with each video shown at different qualities. Only the $\text{CNN}_{M=10}$ videos were used for testing.

Overall Quality. The overall quality Mean Opinion Score (MOS) considers the entire video frame. Without the EL the CNN outperformed bicubic interpolation. For almost all the cases, this remained true even with the EL. Results further showed that CNN $\times 2$ upscaling after lossless compression of the downscaled data results in videos that are comparable to the original.

ROI Quality. The ROI quality measures the quality of the video area that is directly affected by the EL. In this case, the results without the EL are close to those of the overall quality. However, with the EL, all MOS scores, even for $\times 4$ upscaling, are > 3. CNNs still outperform bicubic interpolation. Again, the CNN $\times 2$ lossless results are close to the original video.

Diagnostic Quality. The diagnostic quality indicates of how well diagnosis can be performed on the presented video. These results are very similar to the ROI quality results, further confirming that diagnosis is highly dependent on ROIs.

C. *Compression Ratios*

The Compression Ratio (CR) measures the reduction in the size of the data due to the compression process. 3 separate cases for the compression ratio are discussed.

Firstly, the downscaling operation which takes place prior to HEVC compression reduces the overall data to be compressed. Figure 4(a) shows that, although downscaling by a factor of x reduces the original data by a factor of x^2, the CR does not increase by this same factor. This is attributed to the way HEVC performs compression: reducing the data reduces the redundancies, making HEVC less efficient. In fact, the larger the downscaling factor, the less the redundancies present and hence, the lower the increase in CR.

Figure 4(b) summarises the CR results when considering only EL compression. Only the results for $M = 10$ are shown, since varying M had relatively no effect on the CR. The ELs are compressed losslessly and thus, the compression that can be obtained is limited. Furthermore, the ELs lack spatial and temporal correlations, further limiting the amount of compression that is achievable. As the QP used to compress the medical video increases, more high frequency content appears in the EL, resulting in less spatial correlations and therefore, a reduction in the CR. This effect also occurs for different scaling factors: the $\times 2$ CRs are better. However, the gain that $\times 2$ ELs have decreases as the QP value increases.

The CRs of both the medical video and the EL together are summarized in Figure 4(c). Lossless compression of the $\times 2$ downscaled video, followed by $\text{CNN}_{M=10}$ upscaling obtains a 28.25% improvement in CR over full resolution lossless compression via HEVC, whilst still having good quality results.

D. *Rate-Distortion Curves*

Figures 5(a) and (b) show the rate-distortion curves for $\times 2$ upscaling methods without and with the EL respectively. As a reference point, the bits per pixel (bpp) for full resolution lossless compression evaluates to 1.6137bpp. Since all M values give very similar results, only $\text{CNN}_{M=10}$ is considered. Only $\times 2$ upscaling methods are shown in the interest of space.

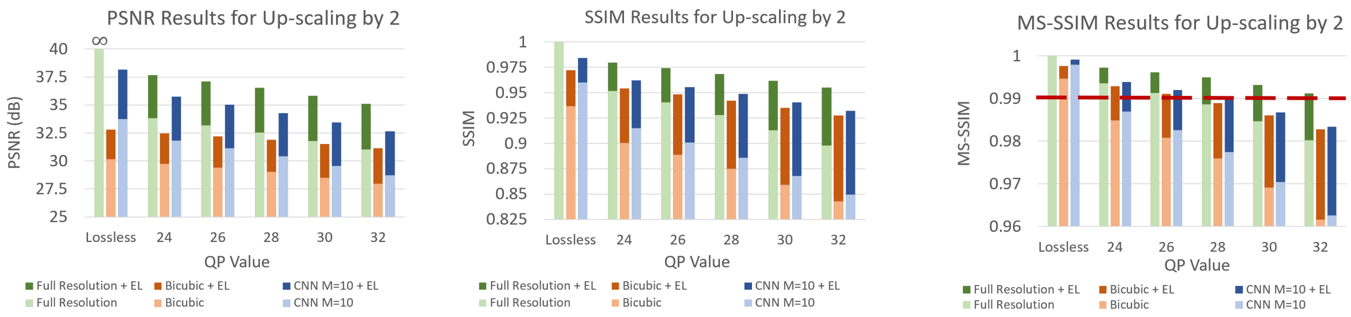
From Figure 5(a), it can be concluded that, at the same bpp, bicubic interpolation has lower PSNR values than the CNN and full resolution methods. Furthermore, it can be observed that for certain bpp values, the CNN method exhibits superior quality compared to full resolution compression with HEVC.

Figure 5(b) shows the improvement in quality that the EL provides, at the cost of a larger bpp. The shape of the curves in both figures is different due to the compression properties of the EL, which vary depending on the quality of the corresponding compressed medical video. Even in this case, bicubic interpolation has lower quality for the same bpp when compared to the other methods. The rate-distortion curves for CNN and full resolution compression overlap at a PSNR of approximately 35.75dB. From Figure 5(b), it can also be seen that the best result of the CNN methods has very good quality (>38dB), yet it is achieved with a lower bpp than full resolution lossless compression (28.25% better).

E. *Timing Results*

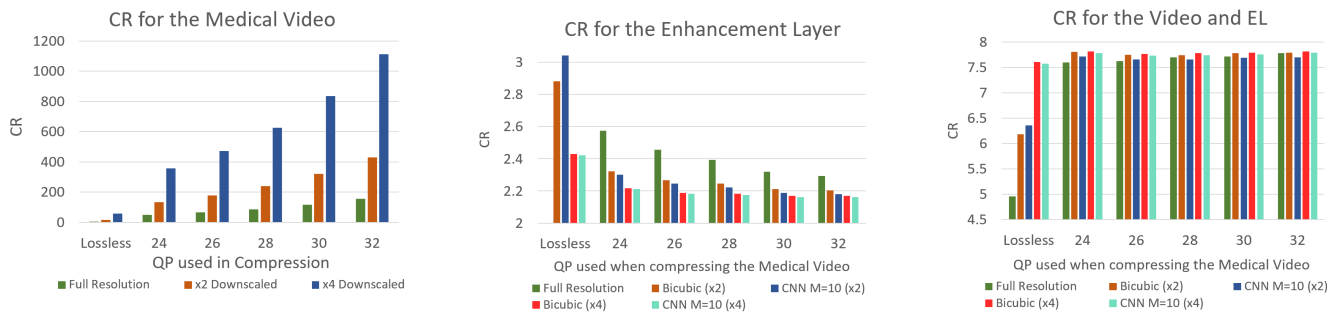
Changes in Encoder and Decoder Timings.

Tables I and II show the total encoding and decoding times in $\%$, where the reference is the time taken to process one full resolution frame losslessly with HEVC ($= 100\%$). Thus, values $< 100\%$ indicate shorter encoding and decoding times while values $> 100\%$ show longer processing times. Timings were obtained using an Intel Core i7-960 processor, clocked at 3.20GHz, with 8GB of DDR3 memory, operating at 1333MHz.



(a) PSNR (b) SSIM (c) MS-SSIM

Fig. 3: Results for $\times 2$ upscaling



(a) Just the Medical Video (b) Just the EL (c) The Medical Video and EL

Fig. 4: Compression Ratios

TABLE I: Timing Results for Encoding. Results shown in %

	Encoding	Up-Scaling	EL Comp.	Total
Bicubic $\times 2$	25.47	0.02	39.61	65.10
CNN$_{M=10}$ $\times 2$	25.47	0.95	39.58	66.00
Bicubic $\times 4$	6.30	0.02	40.57	46.89
CNN$_{M=10}$ $\times 4$	6.30	0.63	40.90	47.83

TABLE II: Timing Results for Decoding. Results shown in %

	Decoding	Up-Scaling	EL Decomp.	Total
Bicubic $\times 2$	27.59	5.23	40.79	73.61
CNN$_{M=10}$ $\times 2$	27.59	204.79	38.89	271.27
Bicubic $\times 4$	9.09	5.30	45.70	60.09
CNN$_{M=10}$ $\times 4$	9.09	134.84	45.48	189.41

An Nvidia Titan Xp with 12GB of GDDR5x memory was also used to run the CNNs. Ubuntu 16.04 LTS was used. As reference, the time taken to encode one frame losslessly is 5.2381 seconds, whilst decoding time is 0.0243 seconds.

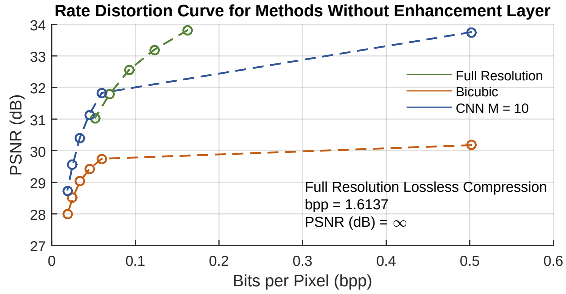
Whilst downscaling reduces the overall encoding time, decoding time is only reduced when bicubic interpolation is used. This is because, since HEVC is asymmetric, decoding time is much faster. Thus, the time gained in decoding the smaller data does not offset the time that is required by the CNN. Downscaling $\times 4$ manages to reduce the video encoding and decoding times by a larger factor. Upscaling via bicubic interpolation is faster than using the CNN, especially when the MC step is considered.

Changes in CNN Timings. Since, up to the deconvolution layer, the CNN operates on the input data size, varying the upscaling factor will also vary the CNN speed. Ignoring MC, the CNN runs approximately 30% faster for $\times 4$ than for $\times 2$ upscaling. The time difference is not exactly equal to the difference in the amount of data due to the data transfers between the host and the GPU. Considering MC, the improvement in runtime becomes approximately 65%: with less input data, MC also runs faster. No speed-up is noticed in bicubic interpolation by changing the upscaling factor.

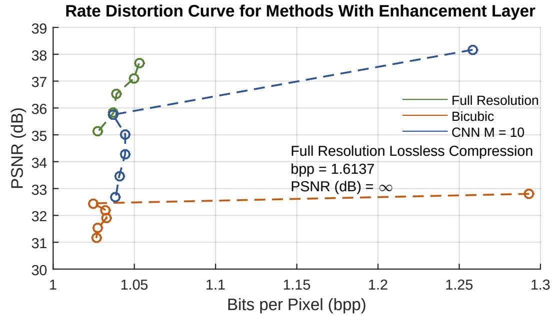
Varying the value of M in the CNNs has very little effect on runtime due to the high parallelization of the GPU. Using a larger M means the CNN has more layers, and thus it takes longer to run. Comparing with $M = 10$, for $\times 2$ upscaling, the difference with $M = 6$ is 0.956ms, whilst with $M = 8$, it is 0.517ms. Similarly, for $\times 4$ upscaling, the differences are 0.864ms and 0.474ms for $M = 6$ and $M = 8$ respectively.

F. Comparison with Other Works

Table III compares this work with other medical video compression systems. Comparison between systems is difficult, especially since different datasets are used. Whilst in this work CT, MRI and Ultrasound scans are considered, most works stick to one or two modalities. In fact, the lossless compression achieved by HEVC in [3], [4] and [6] is 1.470bpp, 1.471bpp and 1.645bpp respectively, whilst in this work this evaluates to 1.614bpp. Results for medical video which was downscaled by 2 and upscaled via CNN$_{M=10}$, with the EL included, are shown, since this system provides the best results.



(a) Without EL



(b) With EL

Fig. 5: Rate-Distortion curves for Full Resolution and $\times 2$ upscaling via Bicubic Interpolation and $\text{CNN}_{M=10}$

TABLE III: Comparison between this work and other literature

Method	bpp	PSNR (dB)	SSIM
Guarda *et al.* [3] - 8 bit dataset	1.267	∞	1
Santos *et al.* [4] - MRP Inter	1.193	∞	1
Santos *et al.* [4] - MRP Inter Diff	1.043	∞	1
Song *et al.* [5] - Lossless	2.986	∞	1
Song *et al.* [5] - PAE	2.42	59.21	/
Song *et al.* [5] - PAE = 8	1.95	47.31	/
Chandrika *et al.* [6] - PLIC-1	1.602	52.53	1
Chandrika *et al.* [6] - PLIC-2	1.598	52.67	1
This Work (EL) - $\text{CNN}_{M=10}$, $\times 2$	1.258	38.17	0.98

In terms of bpp, the proposed method outperforms all the others, except for [4]. The other methods tend to have higher objective quality metrics. This is in fact one of the limitations of this comparison: whilst this work focuses on ROI quality, other compression systems focus on overall quality. However, since diagnosis is still possible, the proposed system provides a good compromise between compression and quality.

VI. Conclusion

In this work, a diagnostically lossless medical video compression scheme is proposed, based on SR via a CNN. The diagnostic quality of the videos is improved by transmitting an EL alongside the video, so that the quality in diagnostic ROIs may be improved. The results show that the compression system can obtain good objective quality metrics whilst maintaining the diagnostic properties of the video. An improvement in CR over HEVC lossless by 28.25% was observed whilst a reduction in the encoding time was also noted.

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